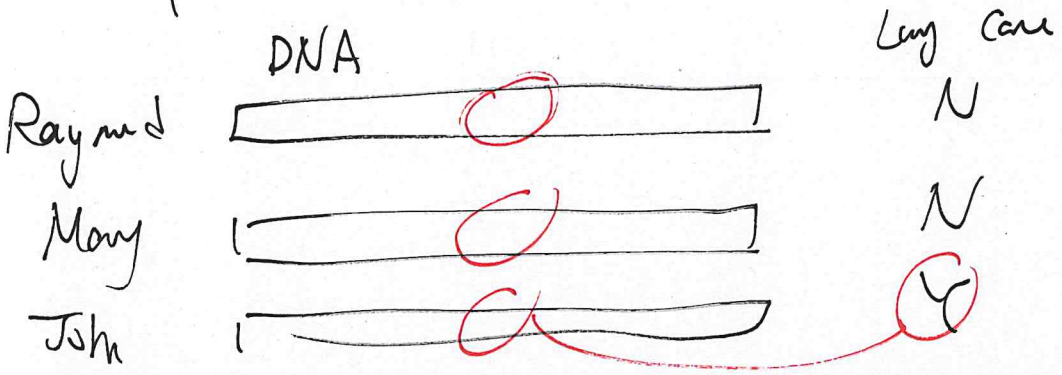
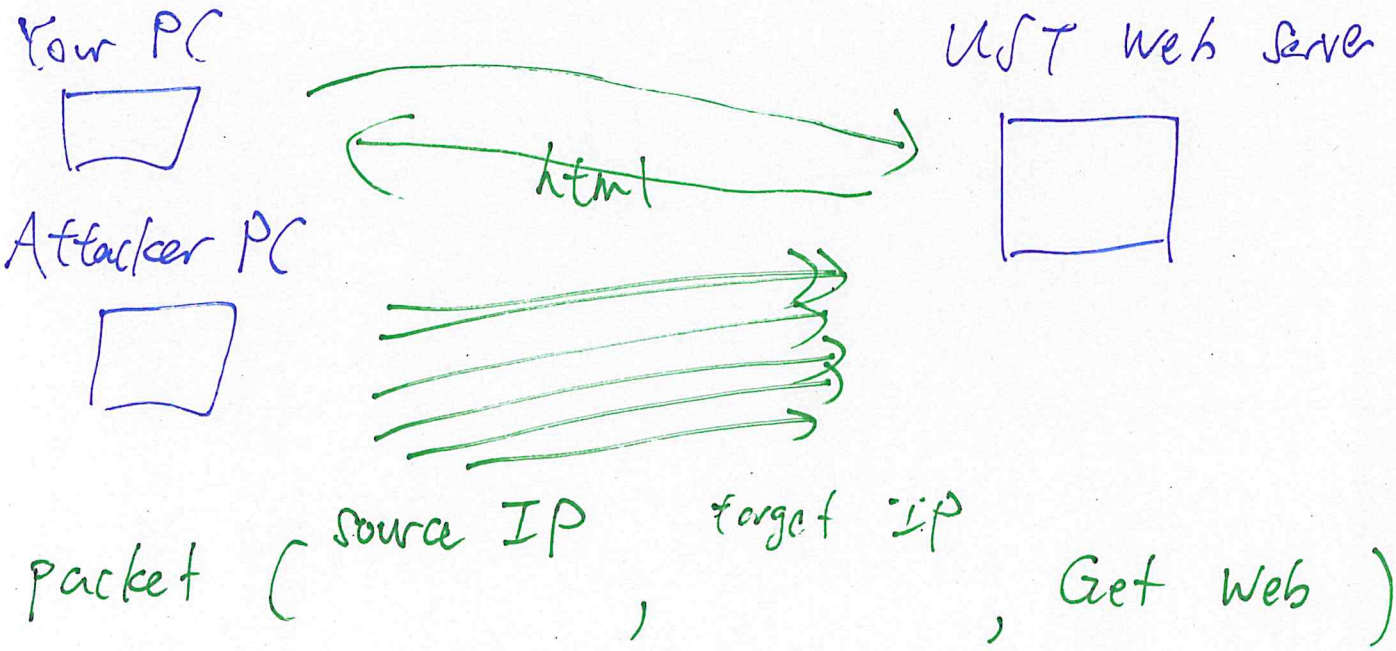
	Gender	Smoker?	Lung Cancer
Raymond	M	N	N
Mary	F	N	N
John	M	Y	Y





$$B \to C \begin{cases} supp(B \to C) = supp(\{B, C\}) \\ conf(B \to C) = \dfrac{supp(\{B, C\})}{supp(\{B\})} \end{cases}$$

if $supp(\{B, C\}) \geq 3$

$supp(\{B\}) \geq 3$?

if $supp([B,C]) \geq 3$,

$supp([B]) \geq 3$ ⊘

B	C
1	1
1	1
1	1
.	.

Goal: to find all rules with supp ≥ 3 and conf. $\geq 50\%$

A	B	C
1	1	1
0	1	1
0	1	1
1	1	0

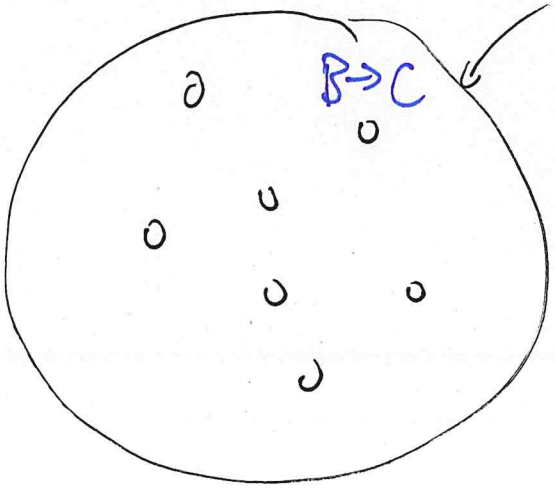
Naive / Straightforward Approach

Step 1: to enumerate all poss. rules

Step 2: to compute the supp. and conf of each rule

Step 3: to extract all rules with supp ≥ 3 and conf. $\geq 50\%$ as "interesting" rules

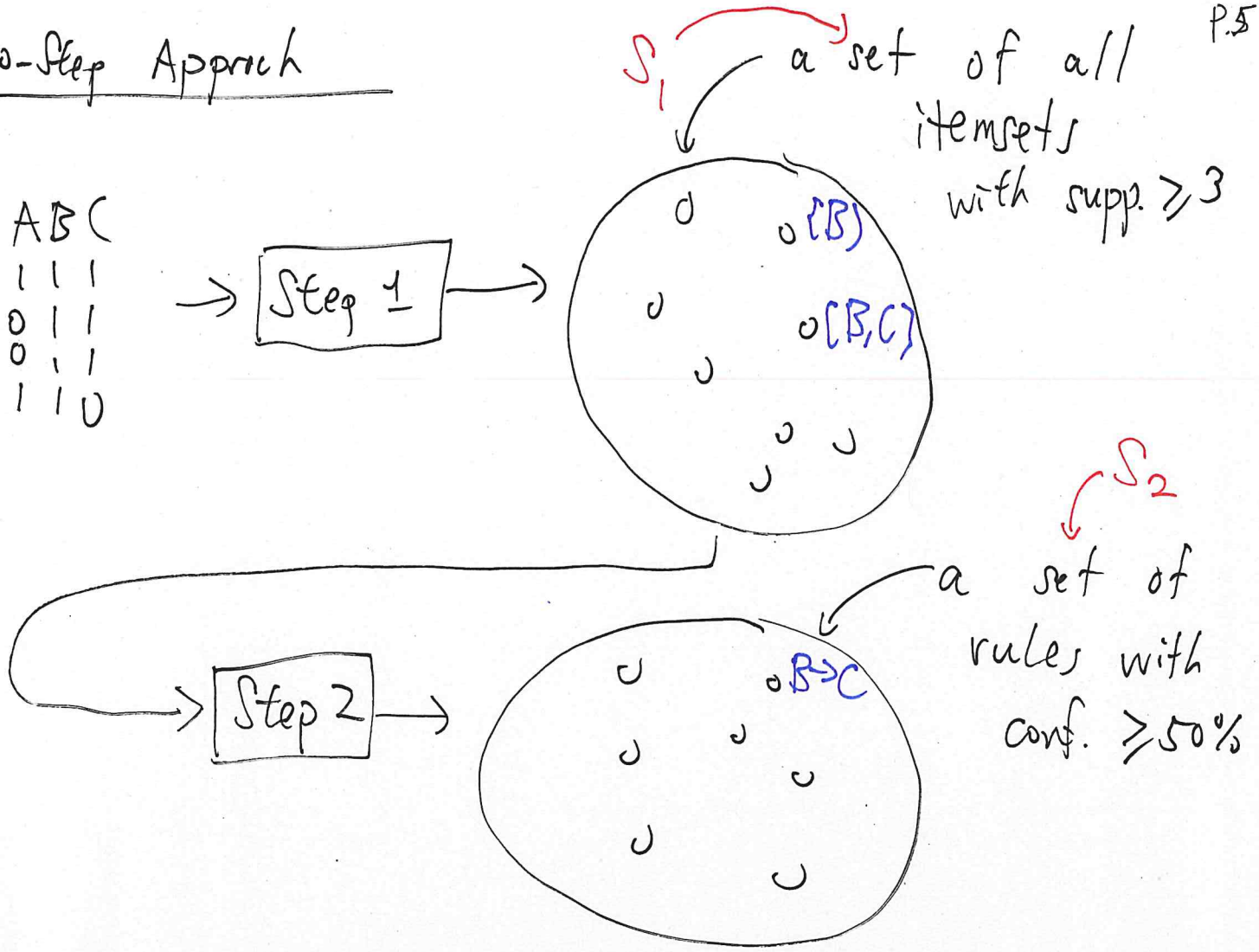
rule	supp.	conf.	interesting?
A→B	2 ✗	100%	N
A→C	1 ✗	50%	N
B→A	1 ✗	50%	N
B→C	3 ✓	75% ✓	Y
C→A	⋮	⋮	
C→B			Y
[A,B]→C			
⋮			



S_0: a set of all rules with supp ≥ 3 and conf. $\geq 50\%$

↑ this set is what we want

Two-Step Approach



Step 2

e.g. {B} {B,C}

- for any 2 elts in S_1, namely X and Y, where $X \subset Y$,
  - if $\frac{supp(Y)}{supp(X)} \geq 50\%$, ——— (*)
    - generate a rule "$X \rightarrow Y - X$"

$B \rightarrow C$

$\frac{supp(\{B,C\})}{supp(\{B\})}$

$Conf(B \rightarrow C)$

e.g. subset

$A = \{X, Y, Z\}$ $C = (X, Y, Z)$

$B = \{X, Y\}$

$B \subset A$ ← subset

$A = C$

$A \subseteq C$

~~$A \Rightarrow$~~

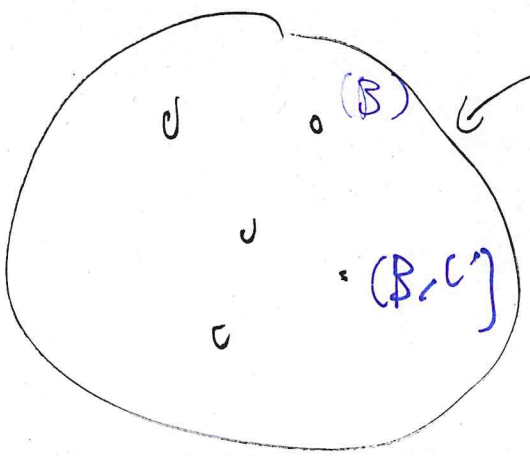
$A - B = \{Z\}$

set minus

Goal: to find all itemsets with supp. ≥ 3

Naive / Stupid first approach

itemset	supp	large?
{A}	2	N
{B}	4	Y
{C}		
{A,B}		
{A,C}		
{B,C}	3	Y
{A,B,C}		



a set of all itemsets with supp. ≥ 3

Show that

$$S_2 = S_0$$

① Claim 1:

$$S_2 \subseteq S_0$$

For each rule in S_2, it is in S_0

② Claim 2:

$$S_0 \subseteq S_2$$

For each rule in S_0, it is in S_2

Claim 1 For each rule in S_2 (e.g. $B \to C$), it is in S_0

- Since $B \to C$ is in S_2
  $conf(B \to C) \geq 50\%$
- The remany question is whether
  $B \to C$ (in S_2) has its supp. ≥ 3

- Since $B \to C$ is in S_2,
  we know that
  we have to calculate $\frac{supp([B,C])}{supp([B])}$
  in step (4)
- In other words,
  $[B,C]$ and $[B]$ are in S_1,
  which means that
  $supp([B,C]) \geq 3$
  and $supp([B]) \geq 3$
- Since $supp(B \to C) = supp([B,C])$
  $supp(B \to C) \geq 3$

Claim 2 For each rule in S_0, (e.g. $B \to C$) it is in S_2

- Since $B \to C$ is in S_0,
  $conf(B \to C) \geq 50\%$ $\Rightarrow$ $\frac{supp([B,C])}{supp([B])} \geq 50\%$
  $supp(B \to C) \geq 3$

- Since $supp([B,C]) = supp(B \to C)$,
  $supp([B,C]) \geq 3$
  Thus, $[B,C]$ is in S_1 ←
- Since $supp([B,C]) \geq 3$,
  $supp([B]) \geq 3$
  Thus, $[B]$ is in S_1 ←

- Since $[B]$ is in S_1,
  and $[B,C]$ is in S_1,
  Step 2 must consider
  $[B]$ and $[B,C]$ together, and
  must generate $B \to C$
  (since $\frac{supp([B,C])}{supp([B])} \geq 50\%$)
- $B \to C$ is in S_2